



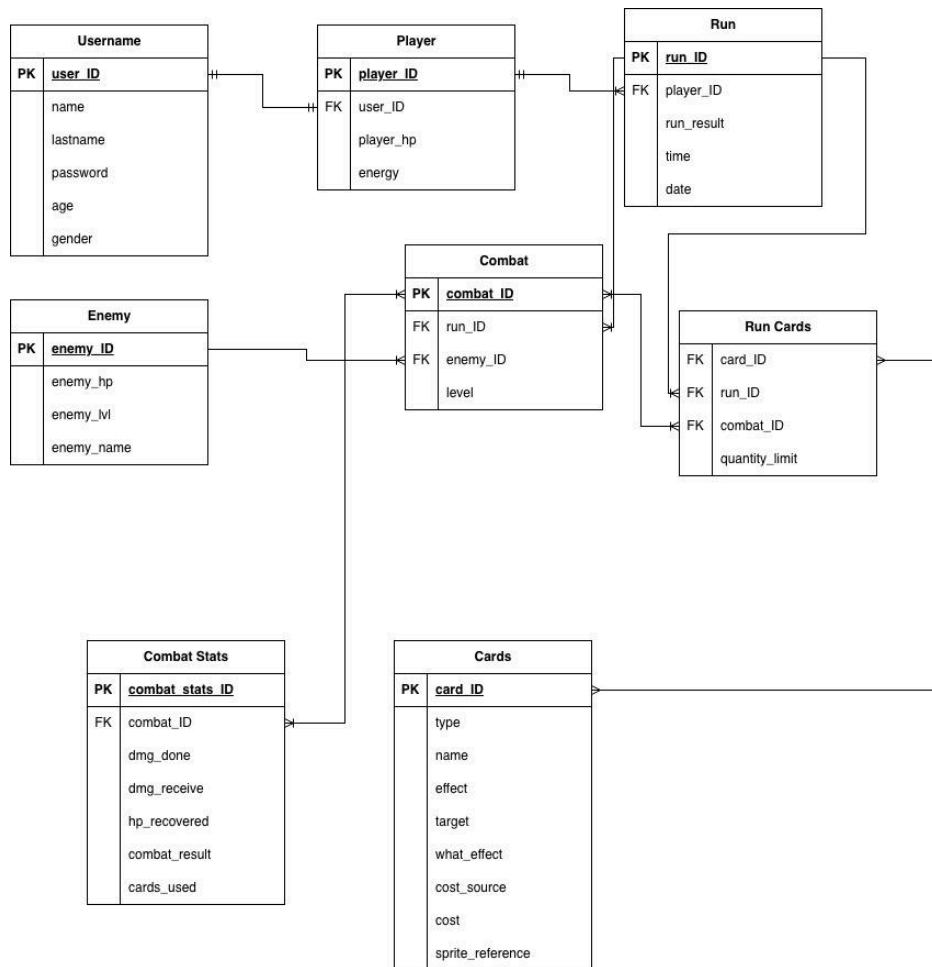
Tecnológico de Monterrey

Database Normalization

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The database design in Catharsis, follows the rules of database normalization to keep data organized and avoid duplication. First, it follows the 1NF because every part of the data is kept in its simplest form and every table has a unique ID (Primary Key) to identify its information. It also meets the 2NF by avoiding that the tables with a single ID have partial dependencies and in more complex tables the extra details depend entirely on the combination of those keys rather than a single one.

The data structure also meets the 3NF by making sure that tables only store information directly related to their main purpose. For example, the “Combat” table only links to an enemy_ID instead of repeating the enemy’s data every time a combat occurs in the game. This separation prevents data mistakes and errors, meaning you can update the data in real time like when updating the HP while in combat without accidentally corrupting the game data.

Finally, the “Run Cards” table acts as a crucial bridge designed to handle the player’s card collection and deck logic during a run. It tracks with specific cards a player has found during a particular run and makes them available for upcoming combats. By storing details like the “quantity_limit” (limits the amount of cards a player can find during a run) here instead of in the main “Card” table the system

allows deck and collection configurations to change dynamically from run to run without affecting the database.

By structuring the database this way, the architecture achieves a seamless balance between the data and the gameplay. It ensures the core mechanics of the game remain stable and simple, as the system never has to reload massive amounts of data during a game's run. This normalization design allows the game to scale making it simple to add new cards, enemies or player stats.